1 Oscillations

1.1 What is an oscillator?

An oscillator is a system that has a stable equilibrium state, to which, if the system is perturbed, it tends to return. Sadly, the system always overshoots, and is compelled to try again, always oscillating around the equilibrium state.[1]

[1] At least, if the system is undamped.

Instances of oscillating systems abound, but as we shall see, they all have in common the same underlying mechanisms. Some examples are:

- A mass on the end of a spring;[2]
- a pendulum;
- a layer of air through which a sound wave is passing;
- the output signal from a certain type of electric circuit.

[2] This is a good prototype, which we will refer to often to discover what makes an oscillator tick, so to speak.

Let's take the oscillations of a layer of air through which a sound wave is passing as an example. These can be intercepted by the membrane of a microphone, which generates an electric signal that reproduces the motion of the air, which can recorded and visualized. When the air is in its equilibrium state, the membrane is still. If we use a point at the centre to represent the motion of the membrane, we can take its "silent" position as the equilibrium state. When a sound wave reaches the microphone, the centre of the membrane oscillates. Figure 1.1 shows a plot of the departure from its equilibrium position of the membrane centre point. It turns out to be a complicated function of time, but the oscillation is clear.

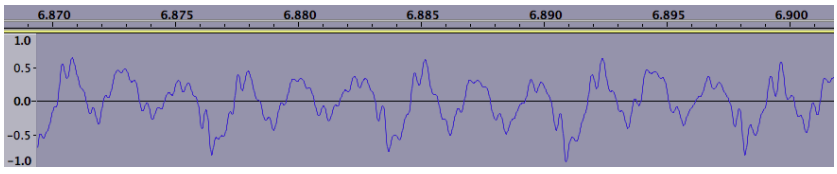


Figure 1.1: Oscillating motion of the membrane of a loudspeaker.

1.2 *Simple Harmonic Oscillations*

A simple harmonic oscillation is a sinusoidal function of time. Figure 1.2 shows an example, when the microphone picks up a pure tone.

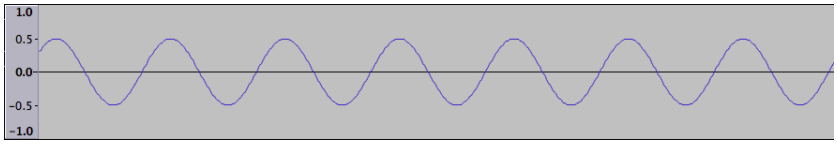


Figure 1.2: A simple harmonic oscillation.

SHO's are important to study, not only because they do arise in practice, but also because they arise all the time! There is a very important theorem due to Joseph Fourier, that guarantees that the most complex oscillations like the one shown in Figure 1.1 can in fact be broken down into a sum of simple harmonic oscillations.

Also, many different oscillations at small enough amplitudes are simple harmonic.

A mathematical description of a SHO

Whatever the nature of the oscillation is – a cork floating on water, the local air pressure at a point where a sound wave is passing, the output voltage or current of an oscillating electric circuit – there is an equilibirum state, and a measurable deviation from that equilibrium state. Let's call the generic deviation from equilbrium $s(t)$. When the oscillation is simple harmonic, $s(t)$ may be written as follows:

$$s(t) = A\cos(\omega t + \phi_0) \tag{1.1}$$

A is the **amplitude**, which we will take to be positive. Since the cosine function oscillates between $+1$ and -1, $s(t)$ oscillates between $+A$ and $-A$.

ω ("omega") is called the **angular frequency** of the oscillation. The meaning of ω is not immediately obvious, but notice that the argument of the cosine function, which must be a dimensionless quantity, is a function of time. We know that the cosine function repeats every 2π radians, and looking at Figure 1.2 we see that $s(t)$ repeats after a certain interval of time, the **period** T. So

$$s(t+T) = A\cos[\omega(t+T)+\phi_0] = s(t) = A\cos(\omega t + \phi_0) \tag{1.2}$$

which means that $\omega T = 2\pi$, or

$$\omega = \frac{2\pi}{T} \tag{1.3}$$

ω is measured in radians per second.

Another term that often comes up is the **frequency** of the oscillation f. This is simply the number of oscillations per second, and a moment's thought will convince you that

$$f = \frac{1}{T} \tag{1.4}$$

The frequency is measured in cycles per second, which is given the name of Hertz (Hz). Do not confuse the frequency f with the *angular* frequency ω! In fact, they differ by a factor 2π.

The last parameter to analyze in equation (1.1) is ϕ_0 ("phi-zero"). The argument of the cosine function is called the **phase** of the oscillation, so it is not surprising that ϕ_0 is called the **initial phase**: At time $t = 0$, the phase is exactly equal to ϕ_0.

Why do we need this constant in the phase? If it wasn't there, the function would be just $s(t) = A\cos(\omega t)$, which at $t = 0$ always equals A. But in general, the oscillation may not start precisely at its maximum value. After all, there's nothing special about $t = 0$; it is simply the time when we choose to start the stopwatch. In fact, if you look at the oscillation of Figure 1.2 again, you'll see that $s(t = 0)$ is not equal to the amplitude. So ϕ_0 fixes that. Its value is determined[3] by whatever the value of s is at $t = 0$.

[3] Albeit only partly, as we shall see.

Example 1.1: *A simple harmonic oscillation has the form*

$$s(t) = (0.50\text{cm})\cos[(100\pi\text{rad}\,\text{s}^{-1})t + 0.75\pi)] \tag{1.5}$$

What are

- *the amplitude,*
- *the frequency,*
- *the period, and*
- *the initial phase*

of the oscillation? After $t = 0$, at what time does the oscillation reach a maximum first?

The amplitude is 0.50cm. The angular frequency ω is $(100\pi\text{rad}\,\text{s}^{-1})$, so the frequency is $f = \omega/2\pi = 50\text{Hz}$, and the period is $T = 1/f = 20\text{ms}$. The initial phase is 0.75π.

The oscillation reaches a maximum when the cosine is equal to 1, that is when the phase is a multiple of 2π. This means that

$$t = \frac{2n - 0.75}{100}\text{s} \tag{1.6}$$

The smallest positive value of t is when $n = 1$, and therefore $t = 12.5\text{ms}$.

1.3 The simple harmonic oscillator equation

If $s(t)$ is the position of the oscillator, then ds/dt is its velocity, and d^2s/dt^2 its acceleration. With $s(t)$ given by equation (1.1), the velocity is

$$\frac{ds}{dt} = -\omega A sin(\omega t + \phi_0) \tag{1.7}$$

and the acceleration is

$$\frac{d^2s}{dt^2} = -\omega^2 A \cos(\omega t + \phi_0 \tag{1.8}$$

so that we can write

$$\frac{d^2s}{dt^2} + \omega^2 s = 0 \tag{1.9}$$

This is the **equation of a simple harmonic oscillator (SHO)**. The fact that the acceleration appears there would seem to indicate that the SHO equation is related to Newton's 2nd. law. In fact, if we multiply by the mass of the oscillator m, this equation tells us that the force on the oscillator is

$$F = -m\omega^2 s \tag{1.10}$$

This type of force is known as an **elastic restoring force**. It is proportional to the negative displacement from equilibrium. So, the greater the displacement in one direction, the stronger the force in the opposite direction, pushing the oscillator back towards equilibrium.

So if for any particular system, we can show that it must satisfy an equation of the form of (1.9), then we know right away (a) that we are in the presence of a SHO, and (b) we can identify the frequency of oscillation as the square root of whatever is multiplying s in the equation. Let's look at a couple of examples.

1.4 Dynamics of a simple harmonic oscillator–examples

A mass on a spring

A spring has one end fixed, and the other end free to move. A mass m is attached to the free end. Measure the position x of the mass from its equilibrium position, positive in the direction in which the spring is stretched[4]. Hooke's law says that the elastic force by the spring on the mass is proportional to x:

$$F = -kx \tag{1.11}$$

This is an elastic restoring force, so the mass moves with simple harmonic motion. Comparing Hooke's law with equation (1.10), we see that the angular velocity is given by

$$\omega = \sqrt{\frac{k}{m}} \tag{1.12}$$

[4] The coordinate x is the perturbation from the equilibrium state, so it plays the role of the variable s in the previous section.

Example 1.2: *When a 250-g mass is suspended from a spring and allowed to oscillate, the period of oscillation is 2.0 s. If a 500-g mass is suspended from the same spring, what is its period of oscillation?*

$$\frac{\omega_{500}}{\omega_{250}} = \sqrt{\frac{k}{500\,\mathrm{g}}}\sqrt{\frac{250\,\mathrm{g}}{k}} = \frac{1}{\sqrt{2}}$$

and $\omega = 2\pi/T$, so

$$T_{500} = \sqrt{2}T_{250} = 2.81\,\mathrm{s}$$

A pendulum

Consider now an ideal pendulum, consisting of a mass m on the end of a string of length L. The equilibrium state is when the string is exactly vertical, so this time the deviation from equilibrium is measured by the angle θ between the string and the vertical direction (Fig. 1.3).

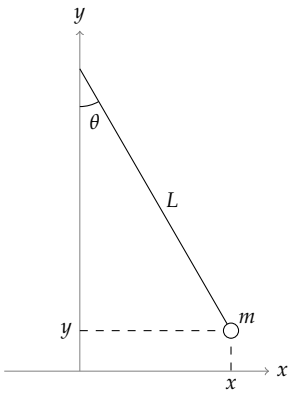


Figure 1.3: An ideal pendulum. The origin of the coordinate system is at the equilibirum position of m.

This time let's start with the principle of conservation of energy. With the origin at the equilibrium position, the coordinates of the mass at any time are

$$x = L\sin\theta \tag{1.13a}$$

$$y = L(1-\cos\theta) \tag{1.13b}$$

Then the kinetic energy is

$$K = \frac{1}{2}m\left[\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2\right] = \frac{1}{2}mL^2\theta^2 \tag{1.14}$$

and the potential energy is

$$U = mgy = mgL[1-\cos(\theta)] \tag{1.15}$$

Conservation of mechanical energy can be expressed as

$$\frac{d}{dt}(K+U) = 0 \tag{1.16}$$

and when we substitute the expressions for K and U, this becomes

$$\frac{d^2\theta}{dt^2} = -\frac{g}{L}\sin(\theta) \tag{1.17}$$

If we restrict ourselves to small-amplitude oscillations (see Appendix B), so that θ is small, then we can approximate $\sin(\theta) \approx \theta$, and therefore

$$\frac{d^2\theta}{dt^2} = -\frac{g}{L}\theta \tag{1.18}$$

Again we have a system that is described by the differential equation (1.9), where θ is the deviation from equilibrium s. Its solution is a simple harmonic oscillation like equation (1.1), if we identify ω with $\sqrt{g/L}$.

1.5 *How to determine A and ϕ_0 from the initial conditions*

We can determine the values of the two unknown constants if we know s, and ds/dt, at $t = 0$. Call these s_0 and v_0. Setting $t = 0$, we find

$$s_0 = A\cos(\phi_0) \tag{1.19a}$$

$$v_0 = -\omega A\sin(\phi_0) \tag{1.19b}$$

The amplitude is

$$A = \sqrt{s_0^2 + \left(\frac{v_0}{\omega}\right)^2} \tag{1.20}$$

and the initial phase is given by

$$\phi_0 = \arctan\left(\frac{v_0}{\omega s_0}\right) \tag{1.21}$$

A word of caution about ϕ_0: There are two possible solutions to equation (1.21)! Your calculator will give you one, and the other is found by adding π to it. So which one is right? It depends on the signs of s_0 and v_0. Say for example that the value of ϕ_0 given by equation (1.21) is in the first quadrant, and therefore the other possible value is in the third quadrant. If the first one is the correct choice, s_0 should be positive, and v_0 should be negative, according to equations (1.19a) and (1.19b). If the second one is correct, then it should be the other way round: s_0 should be negative and v_0 should be positive. So we have to look at the signs of the initial values to pick the correct value of ϕ_0.

1.6 *Alternative representations of a SHO*

The oscillation

$$s(t) = A\cos(\omega t + \phi_0) \tag{1.22}$$

can also be written

$$s(t) = A\sin(\omega t + \psi_0) \tag{1.23}$$

where $\psi_0 = \phi_0 + \pi/2$. Yet another way of writing the same oscillation is

$$s(t) = B\cos\omega t + C\sin\omega t \tag{1.24}$$

If we expand the cosine in equation (1.22), we see that $B = A\cos\phi_0$ and $C = -A\sin\phi_0$.

In these notes, I will generally use expression (1.22), but may occasionally want to use sine instead of cosine, for example, if that is

convenient. This should not be cause for alarm; all three expressions are equivalent, if we choose the constants appropriately.

1.7 Phasors

As we saw at the beginning of our discussion of oscillators, in general an oscillation can be broken down into a sum, or superposition, of simple harmonic oscillations. The problem we want to tackle next is, what does the superposition of several SHM's look like? To answer this question, we will resort to mathematical trickery!

Before proceeding further, please review Appendix A on complex numbers.

Using Euler's identity

$$e^{ix} = \cos(x) + i\sin(x) \tag{1.25}$$

the oscillation $s(t) = A\cos(\omega t + \phi_0)$ can be seen as the real part of a complex function [5]

$$\tilde{s}(t) = Ae^{i(\omega t+\phi_0)} \tag{1.26}$$

Similarly, $s(t) = A\sin(\omega t + \psi_0)$ can be seen as the imaginary part of the same complex function. The function $\tilde{s}(t)$ is called a phasor. It is often convenient to write the part $Ae^{i\phi_0}$ as a single quantity called the complex amplitude $\tilde{A}$. Then the phasor is written as

$$\tilde{s}(t) = \tilde{A}e^{i\omega t} \tag{1.27}$$

The phasor may be visualized as a vector in the complex plane, that is equal to $\tilde{A}$ at $t = 0$, as shown in Figure 1.4, and rotates with angular velocity ω, as it's multiplied by $e^{i\omega t}$.

[5] In general, a tilde on top of a letter will mean that the variable is complex, and therefore has both a real part and an imaginary part.

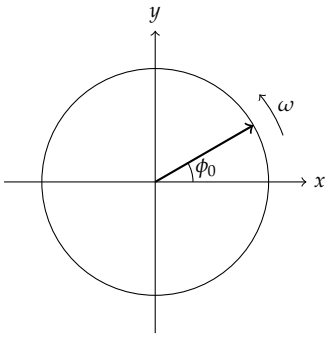


Figure 1.4: A phasor in the complex plane at $t = 0$.

Superposition

To superpose several SHO's, we think of them first as the real parts of some complex phasors, then we will add the phasors, and finally take the real part of the result[6]. This may seem an unnecessary complication, but in fact it's quite the opposite, as we shall see.

[6] Or we can equally think of the SHO's as the imaginary parts of some complex phasors, and at the end take the imaginary part of the result.

Let's work with an example: We will add the oscillations

$$s_1(t) = A\cos(\omega t + \alpha) \tag{1.28a}$$

$$s_2(t) = B\cos(\omega t + \beta) \tag{1.28b}$$

Since they are both cosine functions, it's best to think of them as the real parts of the complex phasors

$$\tilde{s}_1(t) = Ae^{i(\omega t+\alpha)} \tag{1.29a}$$

$$\tilde{s}_2(t) = Be^{i(\omega t+\beta)} \tag{1.29b}$$

Next, we will absorb the initial phases into the amplitudes, creating complex amplitudes

$$\tilde{A} = Ae^{i\alpha} \tag{1.30a}$$
$$\tilde{B} = Be^{i\beta} \tag{1.30b}$$

so the phasors can now be written more simply:

$$\tilde{s}_1 = \tilde{A}e^{i\omega t} \tag{1.31a}$$
$$\tilde{s}_2 = \tilde{B}e^{i\omega t} \tag{1.31b}$$

Now it is clear how to add them:

$$\tilde{s}(t) = \tilde{s}_1 + \tilde{s}_2 = (\tilde{A} + \tilde{B})e^{i\omega t} \tag{1.32}$$

So the superposition has a complex amplitude $\tilde{C} = \tilde{A} + \tilde{B}$. Writing $\tilde{C} = |\tilde{C}|e^{i\gamma}$, we can identify the magnitude $|\tilde{C}|$ with the amplitude of $s(t)$, and γ with the initial phase of $s(t)$.

Let's work through an example with numbers: Let $A = 1.00$ cm, $B = 2.00$ cm, $\alpha = \pi/4$, and $\beta = 2\pi/3$.[7]

[7] In numerical examples, I will give the answer to 3 sig. figs., and work with an extra sig. fig. to avoid rounding errors.

To add the complex amplitudes, it's best to write them in Cartesian coordinates:

$$\tilde{A} = A\cos\alpha + iA\sin\alpha = (0.7071\text{ cm}) + i(0.7071\text{ cm}) \tag{1.33a}$$
$$\tilde{B} = B\cos\beta + iB\sin\beta = (-1.000\text{ cm}) + i(1.732\text{ cm}) \tag{1.33b}$$

Now we add to get $\tilde{C}$:

$$\tilde{C} = (-0.2929\text{ cm}) + i(2.449\text{ cm}) \tag{1.34}$$

The magnitude of $\tilde{C}$ is then

$$|\tilde{C}| = \sqrt{(-0.2929\text{ cm})^2 + (2.449\text{ cm})^2} = 2.466\text{ cm} \tag{1.35}$$

and its angle γ is

$$\gamma = \arctan\left(\frac{2.449\text{ cm}}{-0.2929\text{ cm}}\right) = -1.452\text{ rad or } 1.690\text{ rad} \tag{1.36}$$

The first value is in the fourth quadrant, and the second one is in the second quadrant. To see which is the one we want, we look at the real and imaginary parts of $\tilde{C}$: Since the real part is negative and the imaginary part is positive, we conclude that it must be in the second quadrant. So

$$\gamma = 1.69\text{ rad} \tag{1.37}$$

or 96.8° if you prefer.

We are almost done. Now we need to remember that the oscillations we were adding were cosine functions, which we have treated as the

real parts of complex phasors. After adding the complex phasors, we just need to take the real part of the result to find the superposition $s(t)$:

$$s(t) = (2.47\text{cm})\cos(\omega t + 1.69\text{rad}) \tag{1.38}$$

We can visualize the procedure by drawing the corresponding phasors and their sum in the complex plane. Since the phasors in equations (1.31a) and (1.31b) are both multiplied by $e^{i\omega t}$, so that both rotate at the same rate, we can take a snapshot of the phasors at $t = 0$, and draw the complex amplitudes $\tilde{A}$ and $\tilde{B}$. The superposition can be drawn as the vector sum, using the parallelogram rule for graphical addition of vectors.

Figure 1.5 shows the three phasors.

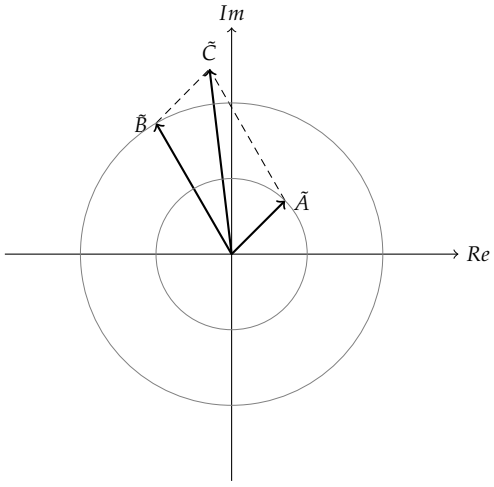


Figure 1.5: Superposition of two complex phasors: $\tilde{C} = \tilde{A} + \tilde{B}$.